MA661E Lab Assignment VT26

Part II: Visualization and Storytelling

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March 11, 2026

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1 Storytelling with Data

1.1 Big Idea

Higher levels of education are strongly correlated with a higher likelihood of earning more than $50K per year in the dataset.

1.2 Three-minute story

The dataset used in this analysis consist of demographic and employment-related information about individuals, including their education level, working hours, and income category. A key question analyzed in this project is whether certain features, such as education level and working hours, are linked with higher income levels.

The first observation comes from the income distribution in the dataset. Most of the individuals fall into the category of earning $50K or less per year, while a smaller proportion earn more than $50K. This imbalance emphasize that higher income levels are less common within the dataset and raises the question of what factors might explain the difference between these two groups.

To explore this further, the relationship between education level and income was analyzed. The visualization comparing education categories with income groups shows a clear pattern: individuals with higher levels of education tend to have a greater proportion of incomes above $50K. For instance, individuals with bachelor's, master's, or professional degrees show a significantly larger share of higher-income outcomes compared to individuals with lower levels of education. However, education levels such as high school or below are more frequently linked with incomes of $50K or less.

Another important factor of the data concerns working hours. On average, individuals earning more than $50K work slightly more hours per week compared to those earning $50K or less. While the difference in working hours is visible, the education variable appears to show a stronger relationship with income in this dataset.

Overall, these findings indicates that education level is an important factor associated with higher income categories in the dataset. While the analysis does not suggest a direct cause and effect, the observed patterns highlight a strong relation between higher educational attainment and an increased likelihood of earning more than $50K.

1.3 Audience

Planned audience for this analysis is students and early-career individuals who are interested in understanding how educational skill relates to income levels. The purpose of this, is to communicate how education may be related with higher income probabilities in the dataset and to present the findings in a clear and accessible way.

1.4 Storyboard

To structure the story, a storyboard was created before assembling the final visuals. The storyboard summary the flow of the study:

- Introduction: Present the income distribution and identify that most individuals earn $50K or less.
- Exploration: Show how education levels relate to income categories using a visualization.
- Supporting information: Compare the average working hours between the two income groups.

- Conclusion: Emphasize that higher educational attainment is strongly linked with a higher likelihood of earning more than $50K in the dataset.

A sketch of the storyboard used to plan the narrative is included below.

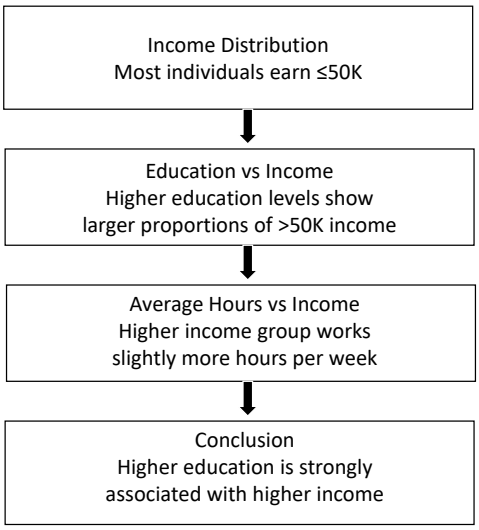

Figure 1: Storyboard illustrating the narrative flow of the analysis.

2 Pseudo-code

1. Load the dataset (CSV) into a Dataframe using pandas.
2. Review the structure of the dataset using functions such as `head()`, `info()`, and `describe()` to understand data types and distributions.
3. Observe that the dataset does not contain column names (columns labeled 0–14) and manually allot meaningful column names such as age, education, hours_per_week, and income.
4. Check for missing values, including special characters such as "?". Confirm that no missing values are present.
5. Execute a univariate analysis of the income variable to examine class distribution.
6. Conduct bivariate analysis between income and hours_per_week by calculating group means and visualizing the relationship using a boxplot.
7. Perform an independent samples t-test to decide whether the difference in average working hours between income groups is statistically significant.
8. Analyze the relationship between education level and income using cross-tabulation and percentage comparisons.
9. Visualize the percentage of individuals earning more than 50K through education levels using a bar chart.

10. Examine potential outliers using boxplot and summary statistics (mean, minimum, maximum, quartiles). Evaluate whether extreme values (e.g., very high working hours) are realistic. Because the values appear reasonable in real-world scenarios, no outliers were removed.

11. Summarize main insights from the exploratory data analysis.

12. Save the cleaned dataset as both an Excel file (.xlsx) and a Pickle file (.pkl) for further analysis and visualization.

3 Visualizations

3.1 Income Distribution

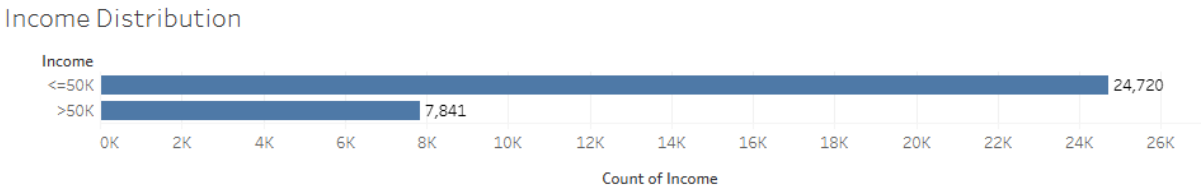

Figure 2: Distribution of individuals across income categories. The majority of individuals earn $50K or less.

The figure shows that most individuals in the dataset fall into the income category of $50K or less, while a smaller portion earns more than $50K. This shows the distribution of income categories is imbalanced in the dataset.

3.2 Education vs Income

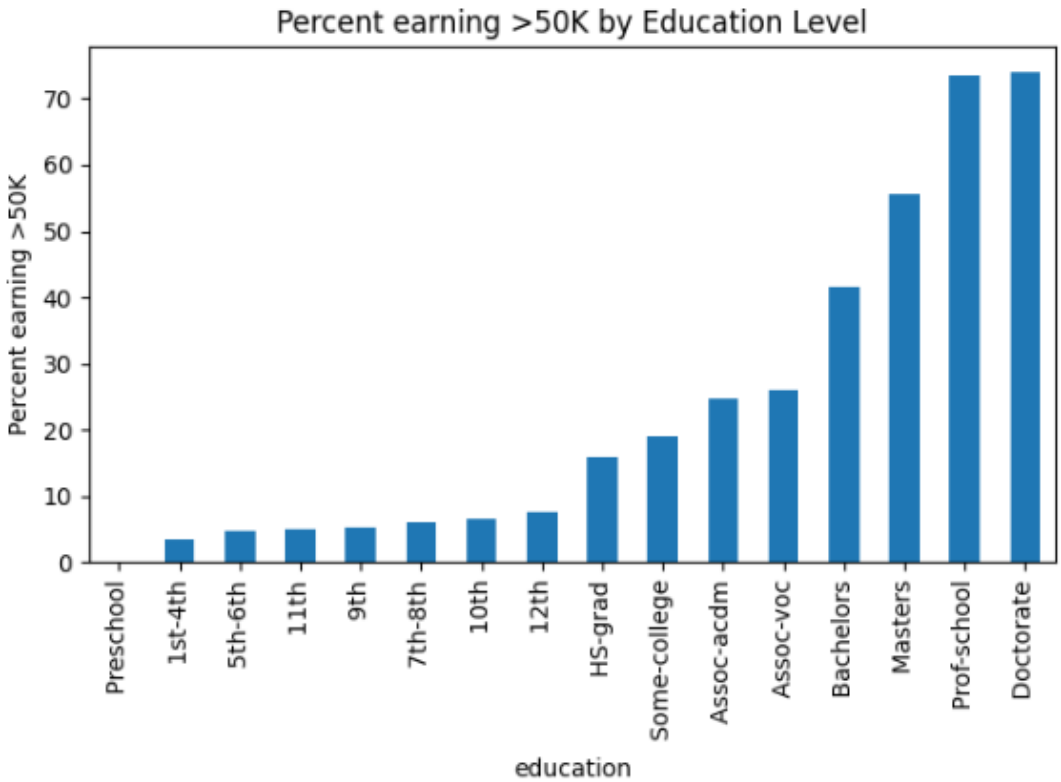

Figure 3: Percentage of individuals earning more than 50K across education levels.

The visualization shows a clear pattern: higher education levels correspond to a higher share of people earning more than $50K. In particular, bachelor's and postgraduate degrees have significantly larger proportions of high-income outcomes compared to lower education levels.

3.3 Average Hours per Week by Income

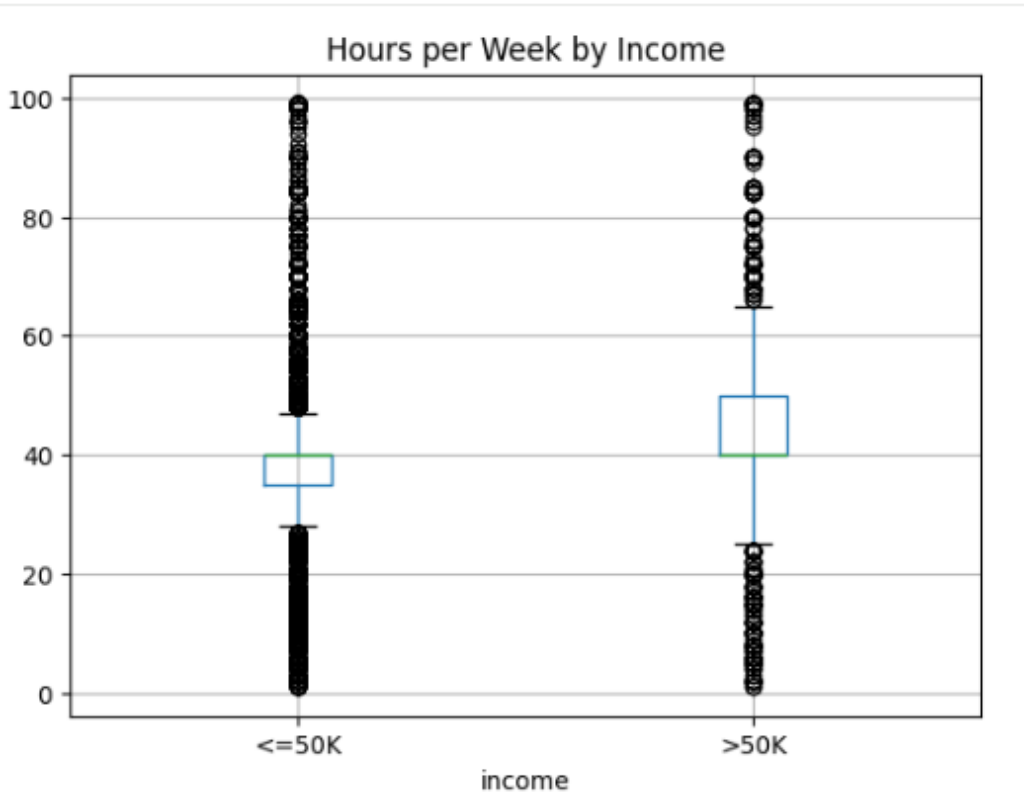

Figure 4: Boxplot showing distribution of working hours by income group.

The boxplot shows that individuals in the ¿$50K income group tend to work more hours per week than those earning $50K or less, as reflected by a higher median and overall distribution. Some extreme values are present in both groups, but they appear plausible and were kept.

3.4 Income Analysis Dashboard (Tableau)

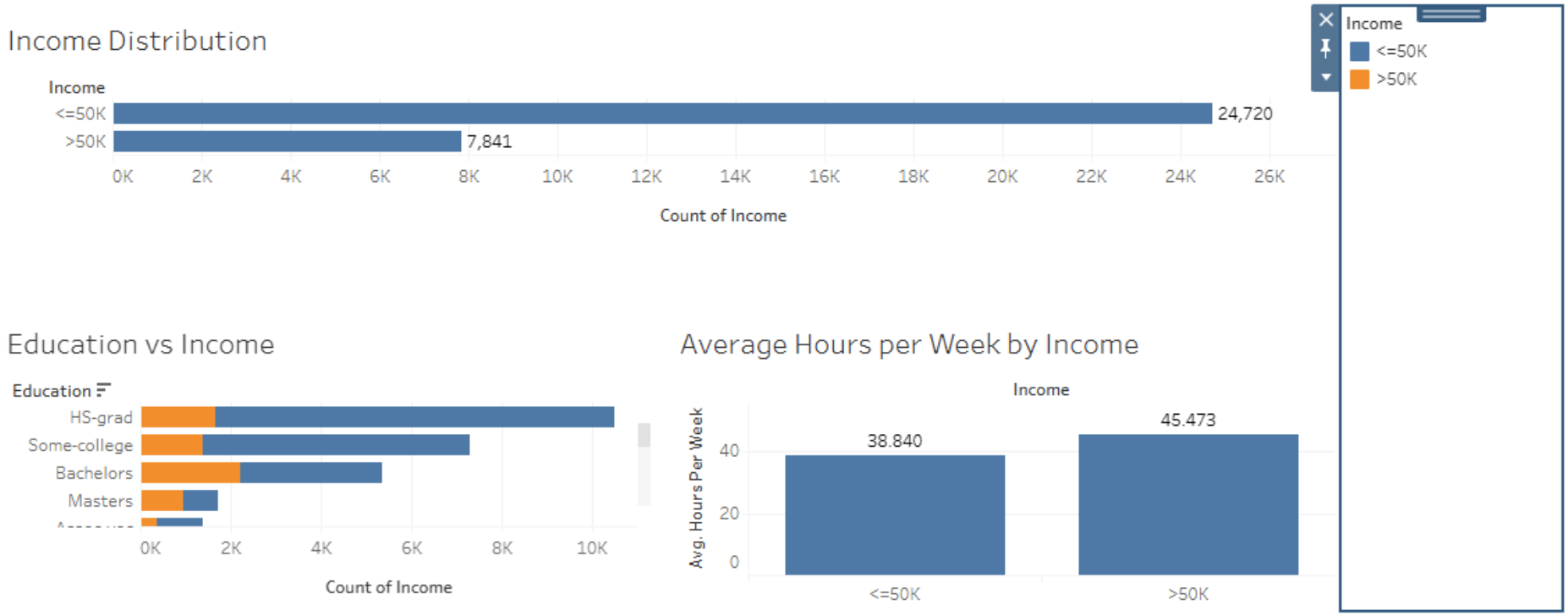

Figure 5: Dashboard created in Tableau to provide an interactive overview of the relationships between income, education level, and working hours.

The dashboard merge key views from the analysis to support interactive exploration of patterns in the dataset. It provides an overview of the income distribution and highlights how education level and working hours relate to income category.

4 Ethical Reflection

4.1 One ethical risk in my story

One ethical risk in this analysis is the possibility of evaluating the relationship between education and income as a causal relationship. The visualizations show that individuals with higher levels of education tend to appear more frequently in the higher income category. However, this does not necessarily mean that education alone causes higher income, since other factors such as experience, occupation, or socioeconomic background may also affect income levels.

4.2 One design decision I changed due to ethics

To reduce the risk of misleading interpretation, visualizations and explanations were carefully structured to highlight association rather than causation. In particular, the story highlights that education is *associated with* a higher probability of earning more than $50K rather than claiming that education directly determines income.

A Appendix

A.1 Exported Tableau Dashboard

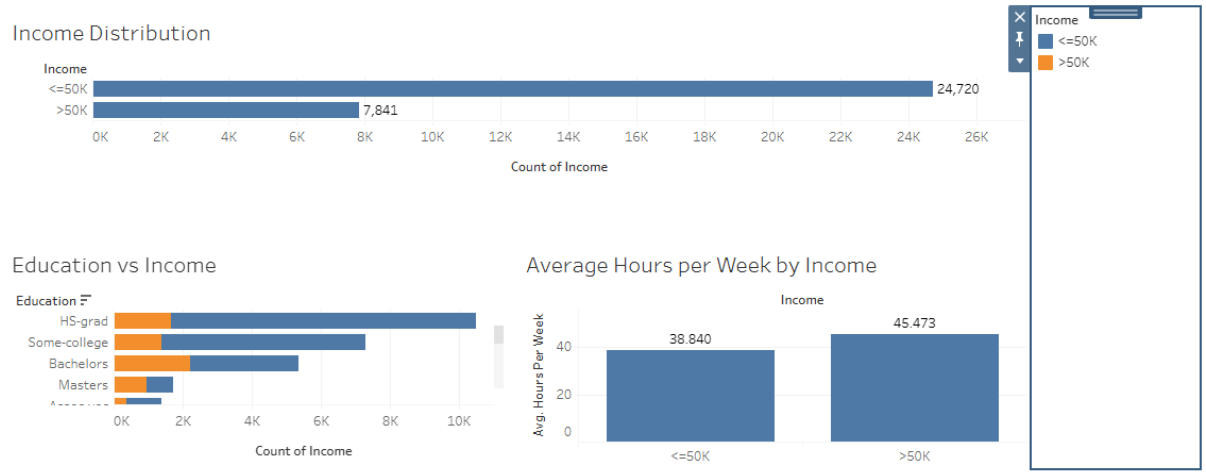

Figure 6: Exported Tableau dashboard summarizing the visual analysis of income distribution, education level, and working hours.